

Nicholas Lee

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Work Experience:

Amazon - Software Development Engineer I

July 2023 - Present

- **Seller Fees Organization** - Systems that configure, execute, and manage all seller fees on Amazon (\$140B+ annually).
- **Tenant Experience team:** Developing a full-stack web app and the Seller Fee configuration store that manages \$25B+ in annual fee changes, releases, approvals, and testing workflows. The configuration store serves as the knowledge base for Execution Services, serving 2.5B+ daily requests.
 - On-call Operations: Participated in on-call rotation for configuration change portals; diagnosed high-severity production issues, implemented systematic fixes to reduce customer contacts, and improved stability.
- **Configuration Dry Run Testing:** Designed and developed a dry-run testing framework for tenant fee configuration updates using AWS Step Functions, Lambda, API Gateway, DynamoDB, and S3 to generate isolated snapshots. Enabled sandbox testing of configuration changes, integrating with Execution Services and Testing platforms.
- **FBA Pan-Europe Oversized Surcharge (Details):** Sole North American engineer coordinating with PMs in Europe and Engineers in China to deliver DSL Configuration and Seller Tool updates (Seller Central web & mobile). Delivered a €0.01 per-unit uplift in FBA PanEU Program, driving +\$23M USD annualized fee revenue in 2024.
- **Seller Fees LLM Slack Bot:** Built a GenAI-powered Slack bot for Seller Fees Q&A with a team of three engineers. Winner of the Seller Fees Hackathon (20+ projects) and Top 10 at the Fulfillment by Amazon GenAI Fair (40+ projects). Leveraged AWS (API Gateway, Lambda, S3, Bedrock Agent).
- **Seller Profile Data Table Migration:** Led migration from a deprecated external table to an in-house table, populated by multiple Apache Spark SQL daily jobs, consolidating 10M+ seller and inventory records for discount calculations.
- **Technologies:** Java, Spring, Angular, AWS, Apache Spark SQL, Domain Specific Languages (DSL)

Amazon - Software Development Engineer Intern

May 2022 - August 2022

- Automated Profitability Management - Supplier Contract Optimization team.
- Designed and built a scalable, reusable email service for Automated Vendor Negotiations, serving 600K+ users.
- Scaled the system to handle 45K+ emails monthly, integrating delivery, click, and rejection metrics.
- Enabled artifact retention for audit and troubleshooting, improving investigation efficiency.
- **Technologies:** Java, Angular, Express, AWS (Lambda, DynamoDB, CloudFormation, CloudWatch)

Arcurve - Software Developer Intern

May 2021 - April 2022

- Contributed to a full-stack web app used by Fortune 500 clients to track and forecast gas delivery data.
- Developed production features and unit tests using AngularJS and C#, including a SQL deployment script that safely modified 200K+ database rows.
- Implemented automated UI tests in C# with Selenium to improve reliability and regression coverage.
- **Technologies:** Angular, TypeScript, Bootstrap, C#, Azure, SQL, CosmosDB, ASP.NET, Selenium UI

Projects:

Portfolio Website: Nick9lee.me

r/Piazza

- Distributed version of r/place (collaborative art canvas), prioritizing replication, fault tolerance and consistency.
- The system utilized two proxies and three servers, each equipped with its own database and automatic restarts.
- Applied timestamps and locks to ensure data consistency across replicated databases.
- Leveraged React, Node.js, MongoDB, AWS EC2, Socket.io, and Nginx.

3D EEG Realtime Data Visualizer - NeuroNexus Hackathon (60 days)

- Developed a Real-time 3D visualization of dynamic brain networks from EEG data using JavaScript and three.js.

Plus 15

- Created an iOS app in Swift to assist users in navigating Calgary's downtown indoor hallways, spanning 16km.
- Integrated Calgary Open Data API to visualize hallway routes and provide lunch spot locations.

Languages: Java, C#, Python, Swift, C++, JavaScript/TypeScript, SQL, HTML/CSS

Technologies: AWS, Azure, Angular, ASP.NET, MongoDB, Apache Spark, Selenium, Git, Hadoop, iOS

Education:

University of Calgary

May 2023

Bachelor of Science in Software Engineering - Graduated with distinction - GPA: 3.61/4.00

Relevant Coursework: DSA, OOP, Software Testing, Databases, Distributed Systems, Data Science, Big Data

Volunteer Work: RoboGals Student Mentor (2022–2023), Calgary Active Habit Association Head Soccer Coach (2017–2019)